

Lab 3: Architectural Justification

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1. Architecture Selection

We chose **Layered Architecture** for the **Vaccination Cohort & Dose Scheduling System**.

Microservices was rejected because splitting dose records, scheduling and certificates into separate services adds network latency, operational overhead and data-consistency risk for a system of this size. Client-Server was rejected as the primary style because a single central server becomes a bottleneck and single point of failure during peak rollout days, and it says nothing about how the business rules inside that server are organised.

2. Two Reasons for the Choice

- **Business rules live in one place.** The system's core rules (cohort eligibility and the minimum-interval check that blocks Dose 2 booking before 28 days, FR-001) all belong in the Business Layer, inside the Vaccination Scheduler. The Presentation Layer (citizen and officer portal) only shows and collects data, and the Data Layer only stores it, so the rules cannot be bypassed or duplicated in the UI.
- **Clear, testable structure with replaceable parts.** The five components map onto three layers and the Scheduler coordinates the other components, so exactly four interfaces are needed. Because layers communicate only through interfaces, the SMS/email provider or the certificate signing method can be replaced without changing the portal or the Scheduler.

3. Security Advantage

The portal can never reach the Database or the Certificate Service directly; every request goes through the Vaccination Scheduler, which authenticates the user (Citizen Registrant or Vaccination Officer), enforces role permissions and validates the request first. Personal health records stay in the Data Layer, reachable only through IDataAccess. Certificates are digitally signed by the Certificate Service, which alone holds the signing key, so a forged or altered QR code fails verification.

4. Performance Benefit

The layers run together as one deployable unit, so calls between components are in-process (milliseconds) rather than network calls between separate services. Certificate verification only needs a signature check on the QR payload, with no database round trip, which keeps it within the 150 ms target in NFR-001 and lets it work offline as well as online. Booking checks read only the citizen's dose history and slot data through indexed queries, so they stay fast under peak load.

5. Component-to-Layer Summary

Component	Layer	Interface (provided by → used by)
User Interface Component	Presentation	Uses ISchedulingService (from Scheduler)
Vaccination Scheduler Component	Business	Provides ISchedulingService; uses ICertificateService, INotificationService, IDataAccess
Certificate Service Component	Business	Provides ICertificateService → Scheduler
Notification Service Component	Business	Provides INotificationService → Scheduler
Database Component	Data	Provides IDataAccess → Scheduler

Vaccination Cohort & Dose Scheduling System - Layered Architecture

UML Component Diagram (5 components · 4 interfaces)

